



THE HISTORY PORTAL | HISTORY DEPARTMENT

EDITION 2026.1

# WAS THE 19TH CENTURY THE REAL TURNING POINT FOR MEDICINE IN BRITAIN?

*Paper 1: Medicine Through Time with the Western  
Front – 18th & 19th C*

Scholar: \_\_\_\_\_

Class: \_\_\_\_\_

### Edexcel Specification: c1700–c1900

- **1. Causes of disease:** From miasma to germ theory (Pasteur, Koch).
- **2. Prevention:** Jenner and the smallpox vaccine; public health reform.
- **3. Treatment:** Simpson and anaesthetics; Lister and antiseptics; Nightingale and nursing.
- **4. Case Study: Cholera and the Broad Street Pump (Snow, 1854).**

## PROGRESS & ASSESSMENT TRACKER

TARGET GRADE: \_\_\_\_\_

| Grade                                                                                        | 9-1 GCSE Grading Scale |       |                  |
|----------------------------------------------------------------------------------------------|------------------------|-------|------------------|
| Lesson / Assessment Title                                                                    | Effort                 | Level | Teacher Comments |
| <b>L1: KT3.1: What breakthrough discoveries changed our understanding of disease causes?</b> |                        |       |                  |
| <b>L2: KT3.2: How did the Industrial Revolution transform prevention and treatment?</b>      |                        |       |                  |
| <b>L3: KT3.3: How significant were Edward Jenner and John Snow?</b>                          |                        |       |                  |
| <b>Final Unit Grade:</b>                                                                     |                        |       |                  |

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## L1: KT3.1: What breakthrough discoveries changed our understanding of disease causes?

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### LEARNING OBJECTIVES

- ☐ Explain the difference between Spontaneous Generation and Germ Theory.
- ☐ Analyze the significance of Robert Koch's scientific methodology in microbiology.

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### Do Now Activity

Score:   / 10

What did William Harvey discover about the human body?

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What did Harvey prove wrong about Galen's theories of blood?

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In what year did the Great Plague hit London?

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What was 'transference'?

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What was the Royal Society?

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Who was Thomas Sydenham?

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Who wrote 'On the Fabric of the Human Body' (1543)?

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What invention helped spread new medical ideas during the Renaissance?

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Why did the influence of the Church begin to decline in the Renaissance?

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What did Vesalius prove about Galen's anatomical ideas?

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Vocabulary Check

**The Odd One Out:** Circle ONE term that does not belong with the others. Explain your reasoning: What historical connection links the other terms, and why is your chosen term different?

Spontaneous Generation | Germ Theory | Bacteriology | Louis Pasteur | Robert Koch | Microscope

Q1. Louis Pasteur and Robert Koch: The Germ Theory

GCSE TWO-SIDED ARGUMENT

GCSE Analytical Assessment: Evaluate the monumental scientific breakthrough of Germ Theory against its delayed clinical application in curing disease.

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| <div>▲ Microbiological Revolution (Change)</div> <ul style="list-style-type: none"><li>Louis Pasteur’s 1861 swan-neck flask experiments decisively disproved the ancient theory of Spontaneous Generation, proving microbes cause decay.</li><li>Robert Koch applied industrial scientific methods: stained specific bacteria with methylene-blue dyes, grew pure cultures on solid agar jelly, and photographed them under high-power microscopes.</li><li>Identified the specific pathogens responsible for deadly diseases: anthrax (1876), tuberculosis (1882), and cholera (1883).</li></ul> <p><i>Starter: “On the one hand, the discovery of Germ Theory was the ultimate turning point in medical understanding because...”</i></p> | <div>■ Delayed Cures &amp; Scientific Resistance</div> <ul style="list-style-type: none"><li>Identifying a bacterial pathogen did NOT mean doctors could cure it; antibiotics like penicillin were not discovered until 1928 and mass-produced until the 1940s.</li><li>British medical authorities fiercely resisted Germ Theory for decades; prominent doctors like Henry Bastian continued arguing for spontaneous generation into the 1880s.</li><li>Patients suffering from tuberculosis and cholera continued to die in huge numbers despite doctors now knowing the microbe responsible.</li></ul> <p><i>Counter: “However, the immediate practical value of Germ Theory was limited because...”</i></p> |
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**GCSE Paragraph Builder: Write a balanced GCSE exam paragraph evaluating the extent to which Germ Theory immediately transformed healthcare in the late 19th century.**

Connectives: While Germ Theory solved the mystery of disease causation... • However, a massive gap remained between diagnosis and cure... • Consequently... • Therefore, its true significance was realized only in...

Q2. Explain how the Enlightenment helped create the conditions for Germ Theory.

Q3. Identify the scientist who published the Germ Theory in 1861.

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#### KNOWLEDGE CHECK (Q4)

**How did Pasteur's swan-neck flask experiment disprove the theory of spontaneous generation?**

- ☐ A. It showed that liquids only decayed when exposed to airborne microbes, proving that microbes caused decay (not the other way around).
- ☐ B. It proved that microbes were killed by intense cold.
- ☐ C. It showed that diseases were spread by physical contact.
- ☐ D. It proved that microbes spontaneously appeared in a vacuum.

**Q5. Describe what Spontaneous Generation was.**

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**Q6. Explain how Robert Koch developed Pasteur's work.**

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**Q7. Evaluate the significance of the Germ Theory on the history of medicine.**

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#### Pair & Share Activity

**Q8. Prompt: Which was more significant in transforming British medicine: Louis Pasteur's theoretical proof of Germ Theory in 1861, or Robert Koch's practical scientific methods to identify specific disease-causing microbes in the 1870s and 1880s?**

*Think: Jot down your choice and one piece of evidence from the sources (e.g., Pasteur disproving spontaneous generation and providing the core principles, vs. Koch growing*

bacteria on agar jelly, staining them with dyes, and finding the TB and cholera microbes).

Your Notes:

16-MARK ESSAY PLANNING MATRIX (2×2 GRID)

Edexcel Paper 1 • Section B (Q5/Q6)

Question: 'Louis Pasteur's publication of the Germ Theory was the biggest turning point in understanding the causes of disease in the period c1700–c1900.' How far do you agree? Explain your answer.

Stimulus: Spontaneous Generation • Robert Koch

✓ Factors Supporting Statement (Agree)

- Pasteur definitively disproved the centuries-old theory of Spontaneous Generation and miasma in 1861, proving via swan-neck flasks that microscopic airborne pathogens caused decay and biological infection.
- Germ Theory provided the foundational scientific principle that transformed medicine, directly inspiring Joseph Lister's antiseptic surgery and compelling governments to construct clean water and sewage systems.

⇔ Counter-Factors / Other Factors (Disagree)

- Pasteur was an industrial chemist whose early work focused on beer, wine, and silkworms; he could not isolate the specific bacteria causing specific human diseases, meaning his theory initially had limited diagnostic value.
- Robert Koch was the decisive practical turning point: he invented solid agar jelly culture, chemical aniline dyes, and photomicrography, systematically isolating the specific pathogens for anthrax (1876), tuberculosis (1882), and cholera (1883).

Level 4 Judgment Criteria: Weigh the relative importance: Was Pasteur's initial theoretical breakthrough the paramount turning point, or was Koch's practical methodology for identifying specific human microbes more transformative?

# GCSE Exam Practice

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**Q9. Describe two features of Robert Koch's scientific methods for identifying bacteria.**

[4 marks • 5 mins]

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**Q10. Explain why Robert Koch's scientific methods were significant in the development of Germ Theory.**

[12 marks • 15 mins]

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**Q11. Explain one way in which ideas about the cause of illness in the period c1700-c1900 were different from ideas about the cause of illness in the period c1500-c1700.**

[4 marks • 5 mins]

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**Q12. 'Louis Pasteur's publication of the Germ Theory was the biggest turning point in understanding the causes of disease.' How far do you agree? Explain your answer.**

**You may use the following in your answer:**

- Spontaneous Generation
- Robert Koch

**You must also use information of your own.**

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## General Notes

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## L2: KT3.2: How did the Industrial Revolution transform prevention and treatment?

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### LEARNING OBJECTIVES

- ☐ Explain how hospital care and nursing standards were transformed c1700-c1900 under the influence of Florence Nightingale.
- ☐ Analyse how the breakthroughs of anaesthetics and antiseptics solved the critical surgical problems of pain and infection.
- ☐ Evaluate the role of government intervention in disease prevention through the 1875 Public Health Act.

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### Do Now Activity

Score:   / 10

What was 'Spontaneous Generation'?

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Who published the Germ Theory in 1861?

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What did Robert Koch contribute to Germ Theory?

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Why did many doctors reject Germ Theory at first?

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What method did Robert Koch use to isolate and photograph bacteria?

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What was 'transference'?

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What did William Harvey discover about the human body?

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Who was Thomas Sydenham?

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What was the Royal Society?

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Who wrote 'On the Fabric of the Human Body' (1543)?

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Vocabulary Check

**The Golden Sentence:** Choose TWO terms from the word bank. Write ONE grammatically sophisticated, historically accurate sentence connecting them using a causal conjunction (because, although, or consequently):

Laissez-faire | Compulsory | Aseptic surgery | Florence Nightingale | Joseph Lister | James Simpson

Q1. James Simpson, Chloroform, and the "Black Period" of Surgery

GCSE TWO-SIDED ARGUMENT

GCSE Analytical Assessment: Evaluate the breakthrough of anaesthesia against the deadly paradox of the "Black Period" of surgery.

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| <p>▲ Conquest of Pain (Change)</p> <ul style="list-style-type: none"><li>James Simpson discovered the anaesthetic properties of chloroform in Edinburgh in 1847, allowing patients to be rendered completely unconscious during surgery.</li><li>Ended the agonizing screams, torture, and physical shock that caused so many surgical patients to die on the operating table.</li><li>Gained royal and societal acceptance after Queen Victoria used chloroform during the birth of Prince Leopold in 1853.</li></ul> <p><i>Starter: "On the one hand, James Simpson's discovery of chloroform transformed surgical practice because..."</i></p> | <p>■ The Deadly "Black Period" Paradox</p> <ul style="list-style-type: none"><li>Paradoxically, death rates from surgery INCREASED in the decades immediately following Simpson's discovery (known as the "Black Period").</li><li>Surgeons operated deeper, longer, and inside the abdomen with unwashed hands and dirty instruments, introducing massive infections.</li><li>Without antiseptics, patients survived the surgery only to die days later from fatal gangrene and sepsis.</li><li>Incorrect dosages of chloroform frequently caused fatal heart failure (e.g. 15-year-old Hannah Greener in 1848).</li></ul> <p><i>Counter: "However, the introduction of chloroform initially worsened surgical mortality because..."</i></p> |
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GCSE Paragraph Builder: Write a balanced GCSE exam paragraph evaluating why the discovery of anaesthesia was initially dangerous for surgical patients.

Connectives: Although chloroform conquered the problem of pain... • Crucially, in the absence of antiseptics... • Consequently... • Therefore, anaesthesia was a dangerous breakthrough until...

KNOWLEDGE CHECK (Q2)

How did Florence Nightingale's 'Notes on Nursing' (1859) help to improve the survival rates of patients in British hospitals?

- ☐ A. It taught nurses how to perform complex internal surgery.
- ☐ B. It established strict protocols for hygiene, ventilation, and cleanliness, drastically reducing infection rates.
- ☐ C. It proved that germs caused disease and taught nurses how to use carbolic acid.
- ☐ D. It taught nurses how to discover new antibiotics.

Q3. Identify the first effective anaesthetic discovered in 1847.

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**Q4. Describe the state of surgery before 1847.**

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**Q5. Explain how Joseph Lister solved the problem of infection.**

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**Q6. To what extent did Simpson's discovery of Chloroform immediately improve surgery?**

### **Pair & Share Activity**

**Q7. Prompt: Was the introduction of anaesthetics (James Simpson's Chloroform) or antiseptics (Joseph Lister's Carbolic Acid) a more significant turning point for patients undergoing surgery in the 19th century?**

*Think: Jot down your choice and one piece of evidence (e.g., Chloroform solved patient pain but initially led to the 'Black Period' of higher death rates, while Carbolic Acid solved postoperative infection and dropped mortality rates from 46% to 15% but relied on Pasteur's Germ Theory).*

**Your Notes:**

## 16-MARK ESSAY PLANNING MATRIX (2×2 GRID)

Edexcel Paper 1 • Section B (Q5/Q6)

**Question:** 'The discovery of Carbolic Acid was the most significant turning point in surgery in the years c1700–c1900.' How far do you agree? Explain your answer.

*Stimulus: Joseph Lister • James Simpson*

### ✓ Factors Supporting Statement (Agree)

- Joseph Lister's introduction of carbolic acid spray (1865) directly overcame the deadly barrier of infection (sepsis and gangrene), reducing surgical mortality in his Glasgow ward from 46% to 15%.
- Lister directly linked Pasteur's Germ Theory to operating theatres, paving the way for late-Victorian aseptic surgery (sterilizing instruments with steam autoclaves, sterile gowns, and rubber gloves).

### ⇌ Counter-Factors / Other Factors (Disagree)

- James Simpson's discovery of chloroform anesthesia in 1847 was the essential prerequisite: by eliminating the barrier of agonizing pain, it allowed surgeons to operate calmly and attempt complex internal operations.
- Early carbolic acid was unpopular and fiercely resisted: it irritated surgeons' skin, and during the 'Black Period' of surgery (1840s–1860s), deeper surgeries under anesthesia without antiseptics had caused infection rates to skyrocket.

**Level 4 Judgment Criteria:** Weigh the relative importance: Was overcoming the barrier of infection with carbolic acid more decisive for patient survival than conquering the barrier of surgical pain with chloroform?



**Q8. Explain why Florence Nightingale's reforms were significant in improving hospital care.**

[12 marks • 15 mins]

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**Q9. Explain one way in which surgical treatments in the period c1700-c1900 were different from surgical treatments in the medieval period (c1250-c1500).**

[4 marks • 5 mins]

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**Q10. 'The discovery of Carbolic Acid was the most significant turning point in surgery in the years c1700-c1900.' How far do you agree? Explain your answer.**

[20 marks • 25 mins]

**You may use the following in your answer:**

- Joseph Lister
- Chloroform

**You must also use information of your own.**

- Joseph Lister
- Chloroform

**You must also use information of your own.**

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## General Notes

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## L3: KT3.3: How significant were Edward Jenner and John Snow?

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### LEARNING OBJECTIVES

- ☐ Explain the significance and limitations of Edward Jenner's development of the smallpox vaccine.
- ☐ Analyze how John Snow investigated the 1854 cholera epidemic and assess the significance of his findings.

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### Do Now Activity

Score:    / 10

Who was Florence Nightingale?

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What did Florence Nightingale believe caused disease?

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Which anesthetic was discovered by James Simpson in 1847?

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What was 'Spontaneous Generation'?

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Who published the Germ Theory in 1861?

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What did Robert Koch contribute to Germ Theory?

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In what year did the Black Death arrive in England?

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What did Vesalius prove about Galen's anatomical ideas?

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Who introduced antiseptic surgery using carbolic acid in 1865?

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What was the significance of the Public Health Act of 1875?

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Vocabulary Check

**Conceptual Classification:** Categorise the terms from the word bank into the two historical boxes below:

|                                                                                      |                          |
|--------------------------------------------------------------------------------------|--------------------------|
| Inoculation   Vaccination   Spot Map   Edward Jenner   John Snow   Broad Street Pump |                          |
| Power, Governance & Warfare                                                          | Economy, Trade & Society |
|                                                                                      |                          |
|                                                                                      |                          |
|                                                                                      |                          |

Q1. Identify the disease Edward Jenner created a vaccine for.

Q2. Describe how Jenner proved his vaccine worked.

Q3. Explain the actions John Snow took during the 1854 Cholera outbreak.

Q4. Evaluate the significance of the 1875 Public Health Act.

KNOWLEDGE CHECK (Q5)

How did John Snow prove that cholera was a waterborne disease in 1854, long before the discovery of germs?

☐ A. By looking at the cholera bacteria under a microscope.

- ☐ B. By mapping the deaths around the Broad Street pump, proving the victims shared a water source rather than breathing the same 'bad air'.
- ☐ C. By boiling all the water in London and watching the disease disappear.
- ☐ D. By injecting a cholera vaccine into local residents.

## Pair & Share Activity

**Q6. Prompt: Whose breakthrough was a more significant turning point in the history of disease prevention: Edward Jenner's 1796 discovery of smallpox vaccination, or John Snow's 1854 discovery that cholera was water-borne?**

*Think: Jot down your choice and one piece of evidence from the sources (e.g., Jenner's vaccine being the first scientific prevention method that eventually wiped out a disease globally, versus Snow proving the link between physical sanitation and disease, which directly paved the way for Bazalgette's sewers and the 1875 Public Health Act).*

### Your Notes:

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## 16-MARK ESSAY PLANNING MATRIX (2×2 GRID)

Edexcel Paper 1 • Section B (Q5/Q6)

**Question:** 'Government action was the main reason for improvements in public health in the second half of the nineteenth century.' How far do you agree? Explain your answer.

*Stimulus: The 1875 Public Health Act • John Snow*

### ✓ Factors Supporting Statement (Agree)

- The government abandoned laissez-faire in favor of state intervention: the landmark **1875 Public Health Act** made sanitary regulations compulsory, requiring councils to provide clean water, collect refuse, and appoint medical officers.
- Parliament funded and authorized massive infrastructure: Joseph Bazalgette's 82-mile London sewer network (1858–1875) following the Great Stink transported 420 million gallons of sewage daily, permanently ending London cholera.

### ⇌ Counter-Factors / Other Factors (Disagree)

- Scientific breakthrough was the underlying catalyst: John Snow's 1854 Broad Street cholera investigation and Pasteur's 1861 Germ Theory provided the undeniable empirical proof that forced reluctant politicians to legislate.
- The expansion of political democracy drove reform: the 1867 Second Reform Act gave working-class urban men the vote, compelling rival political parties (Disraeli's Conservatives) to deliver tangible public health reforms.

**Level 4 Judgment Criteria:** Weigh the relative importance: Was compulsory government legislation the decisive primary cause of public health reform, or was legislation merely the administrative response to scientific proof and democratic voter pressure?



**Q7. Explain why Edward Jenner's smallpox vaccination was significant, despite its limitations.**

[12 marks • 15 mins]

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**Q8. Explain one way in which the role of the government in preventing disease in the period c1700-c1900 was different from the period c1500-c1700.**

[4 marks • 5 mins]

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**Q9. Explain why there was rapid progress in public health in the second half of the nineteenth century (c1850-c1900).**

[12 marks • 15 mins]

**You may use the following in your answer:**

- John Snow
- The Great Stink

**You must also use information of your own.**

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## General Notes

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# End of Unit Reflection & Pupil Voice

*Unit: Paper 1: Medicine Through Time*

**Instructions:** Now that you have completed your final assessment, take 10 minutes to reflect on your learning across this entire unit. Be honest—this is your chance to use your **Pupil Voice** to help your teacher plan future lessons.

## 1. What Went Well (WWW)

What topic or skill did you find most engaging or easiest to understand in this unit?

## 2. Even Better If (EBI)

What did you find most challenging? Are there any concepts you are still confused about?

## 3. Effort & Target Setting

Circle your effort level for this unit: **1 (Low)**   **2**   **3**   **4**   **5 (Excellent)**

What is one specific target you want to set for yourself in the next unit?

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## Teacher Feedback & Coaching

(To be completed during live marking / feedback lessons)

## Factors Overview: 18th & 19th C

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Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

| Factor                           | Specific Historical Example & Impact |
|----------------------------------|--------------------------------------|
| <b>Individuals</b>               |                                      |
| <b>The Church &amp; Religion</b> |                                      |
| <b>Government &amp; Wealth</b>   |                                      |
| <b>Science &amp; Technology</b>  |                                      |
| <b>Attitudes in Society</b>      |                                      |
| <b>War</b>                       |                                      |